

UNIVERSITY OF EASTERN AFRICA ,BARATON

SCHOOL OF BUSINESS

DEPARTMENT OF INFORMATION SYSTEM AND COMPUTING

ASSIGNMENT DONE IN THE PARTIAL FULFILLMENT OF THE COURSE

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COURSE TITLE: SYSTEM ANALYSIS AND DESIGN

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BASELINE PROJECT PLAN (BPP)

DAIRY COLLECTION, QUALITY, AND FARMER PAYMENT MANAGEMENT SYSTEM .

1) INTRODUCTION

1. Clients and Stakeholders

This project is for Sirikwa Dairies and General PLC, a dairy business in Ziwa, Uasin Gishu County. The primary stakeholders are company management, farmers, collection-centre workers, quality-control staff, transport workers, accountants, and IT staff.

2. Project Overview

The proposed system will centralize farmer details, milk deliveries, quality results, and payment information into a single digital platform. Currently, using paper records or separate files makes information retrieval and report generation slow and prone to error. This project aims to streamline record-keeping, reduce errors, and provide managers with better data for decision-making.

3. Recommended Solution

The company should first evaluate existing dairy software. If no suitable and cost-effective option is found, a custom system should be developed. This system will be introduced in stages using an **Agile approach,** allowing staff to test each version and provide feedback before full implementation.

4. Project Scope

a. Register farmers and maintain their contact details.
b. Record milk quantities, delivery dates, and collection centres.
c. Store milk-quality test results.
d. Calculate proposed farmer payments based on approved rates and deductions.
e. Generate daily, weekly, and monthly reports.
f. Provide managers with tools to compare collection data and identify trends.

2) SYSTEM DESCRIPTION

- **Alternative Solutions**

There are three possible ways Sirikwa dairies could address the problem:

1. **Continue with Paper Records and Spreadsheets:** This has a low initial cost but perpetuates slow reporting, manual errors, and difficulty in data retrieval.

2**. Buy an Existing Dairy Management System:** This could save development time but may involve high subscription costs, setup fees, and limited customization to fit Sirikwa Dairies' specific needs.

3**. Develop a Custom System:** This option is tailored to the company's exact operations but requires significant investment in time, money, and skilled personnel for development and testing.

- **Recommended Alternative**

For this academic proposal, the preferred option is a tailored system. It will integrate three core capabilities:

I. **Transaction Processing System (TPS):** To record daily activities like milk deliveries.
II. **Management Information System (MIS):** To generate reports on collections, quality, and payments.
III. **Decision-Support:** To help managers analyze data and plan.

The project will follow an **Agile methodology,** building the system in small, iterative stages. This allows for early problem detection and continuous improvement based on user feedback.

3) FEASIBILITY ASSESSMENT

- **Estimated Project Cost**

Activity	**Estimated cost(ksh)**
Requirements and planning	140,000
System and database development	370,000
Computers, servers and Network Equipment	220,000
Data entry devices and setup	150,00
Security, Backup, and Recovery	100,000
Checking and Transferring Existing Records	90,000

Testing	100,000
Training and User Guides	70,000
Installation and Initial Support	90,000
Unexpected Expenses Allowance	150,000
TOTAL	1,480,000

- **Expected Benefits**

The system is expected to reduce errors in farmer and milk records, speed up report preparation, and help staff verify payment calculations. Managers will be able to compare collection-centre figures and monitor milk quality more efficiently.

- **Technical Feasibility**

The system will use standard computers, a central database, and a web browser, with mobile devices for field use. Key challenges include unreliable internet at some centres, errors in old records, and limited staff experience with technology. These will be addressed through training, data verification, and regular backups. Offline data recording will be considered for areas with poor connectivity.

- **Operational Feasibility**

The system is designed to fit the daily workflows of collection, quality-control, and finance staff. Staff involvement in planning and testing is crucial. The project will be piloted at one collection centre to resolve major issues before a wider rollout.

- **Legal and Security Issues**

The system will contain sensitive personal and company data. The company must comply with the Data Protection Act, 2019. Access will be restricted to authorized personnel. Clear rules will be established for data entry, viewing, correction, and approval, with regular backups to ensure data integrity and security.

- **Project Schedule**

STAGE	**MAIN ACTIVITY**	**DURATION**
Planning	Agree on goals and identify stakeholders	1 week

Requirements	Gather user needs	2 weeks
Design	Plan database and system screens	2 weeks
Development	Build system and database	4 weeks
Testing	Check system and fix errors	2 weeks
Data preparation	Check and transfer approved records	1 week
Training & pilot	Train staff and test at one center	2 weeks
Final review	Fix remaining issues and assess wider use	2 weeks
		Total-16 weeks

4) MANAGEMENT ISSUES

- **Management Concerns**

✓ **Cost Control:** Monitor spending and approve major extra costs.
✓ **Staff Training:** Provide practical training and support for staff.
✓ **Record Accuracy:** Verify milk quantities and farmer details.
✓ **Security:** Protect farmer information from unauthorized access.
✓ **System Failure:** Maintain backups and a recovery plan.
✓ **Time Management:** Review progress regularly and report delays early.

- **Project Team**

Team Role **Main Responsibility**

1.Project Sponsor: Approves project and major decisions

2.Project Manager: Manages work, schedule, budget, and risks

3.Systems Analyst: Gathers and documents user requirements

4.Software Developer: Builds the system

5.Database Administrator: Manages and protects stored data

6.Tester: Checks the system and reports errors

7.Operations/Finance Reps: Explain business rules and validate reports

8.IT Support & End Users: Support equipment, test, and provide feedback

In a small team, roles may overlap, but approvals and access to sensitive data must remain properly controlled.

- **Communication Plan**
  - **Weekly Team Meetings:** Project team members.
  - **Weekly Progress Reports:** Project manager to sponsor.
  - **System Demonstrations:** Developers to users during development.
  - **Testing Meetings:** Users, testers, and developers.

- **Standards and Procedures**

The team will use written and approved requirements, track changes, validate data before entry, enforce access controls, test before launch, perform regular backups, and use simple user guides.

- **Main Risks and Responses**

Schedule/cost: overrun Weekly progress and spending reviews

Incorrect old records: Check and correct records before transfer

Staff struggle with system: Provide training and support

Internet interruption: Check connectivity and assess offline options

Lost information: Keep backups and test recovery

Incorrect payment results: Test calculations and check before payment

Extra features delay work: Keep to agreed scope and approve changes formally

5) CONCLUSION

The proposed system aims to help Sirikwa Dairies improve the accuracy and efficiency of its milk collection, quality, and payment processes. With an estimated budget of KSh 1,480,000 and a 16-week timeline, the system will be developed in stages using an Agile

approach. Before implementation, the company should confirm its needs, costs, and evaluate any existing software options.

6) REFERENCES

- Sirikwa Dairies and General PLC. (n.d.). Official website. https://sirikwadairies.co.ke/·
- Republic of Kenya. (2019). Data Protection Act,2019

7) GITHUB REPOSITORY

https://github.com/KELVIN2-afk/Sirikwa-dairies-management-system-